



User Manual **RFS-1000**

Signal Generator

WARRANTY

All Berkeley Nucleonics (BNC) instruments are covered by a 1 year return-to-manufacturer (90 days high current/high voltage pulsers) repair warranty for parts and labor from the date shipment against defective materials and/or workmanship provided the product has been subjected to proper use, handling and storage. Berkeley Nucleonics shall not be responsible for defects caused by natural wear and tear, willful damage, negligence, force majeure, and abnormal working conditions or failure to follow Berkeley Nucleonics spoken or written guidelines as to the storage, installation, commissioning, use or maintenance of the products or (if there are none) good trade practice; or the customer alters or repairs such goods without written consent from Berkeley Nucleonics. Opening the instrument and/or breaking warranty seals voids the warranty. Service plans are subject to the same warranty policy regarding use.

IMPORTANT! PLEASE READ CAREFULLY

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General Warranty

BERKELEY NUCLEONICS warrants that the product will be free from defects in materials and workmanship for 1 year from the date of purchase of the product by the original purchaser from the BERKELEY NUCLEONICS Company. The warranty period of accessories such as probe, battery, adapter is 12 months. This warranty only applies to the original purchaser and is not transferable to the third party.

If the product proves defective during the warranty period, BERKELEY NUCLEONICS either will repair the defective product without charge for parts and labor or will provide a replacement in exchange for the defective product. Parts, modules and replacement products used by BERKELEY NUCLEONICS for warranty work may be new or reconditioned like new performance. All replaced parts, modules and products become the property of BERKELEY NUCLEONICS.

To obtain service under this warranty, Customer must notify BERKELEY NUCLEONICS of the defect before the expiration of the warranty period. Customer shall be responsible for packaging and shipping the defective product to the service center designated by BERKELEY NUCLEONICS, and with a copy of customer proof of purchase.

This warranty shall not apply to any defect, failure or damage caused by improper use or improper or inadequate maintenance and care. BERKELEY NUCLEONICS shall not be obligated to furnish service under this warranty a) to repair damage resulting from attempts by personnel other than BERKELEY NUCLEONICS representatives to install, repair or service the product; b) to repair damage resulting from improper use or connection to incompatible equipment; c) to repair any damage or malfunction caused by the use of non-BERKELEY NUCLEONICS supplies; or d) to service a product that has been modified or integrated with other products when the effect of such modification or integration increases the time or difficulty of servicing the product.

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GENERAL REMARKS

The RFS-1000 family comprises two CW and sweep RF/microwave generators—**RFS-1220** (0.1 – 22 GHz) and **RFS-1420** (0.1 – 42 GHz)—each delivering up to **+15 dBm** output. They're engineered for automated test, impedance matching systems, antenna development, aerospace & defense research, wireless infrastructure build-out, and satellite-link validation. With straightforward controls, broad frequency coverage, and a rugged aluminum chassis, they deliver reliable performance in both lab and production environments.

STAND-ALONE USE QUICK SETUP

Connect your device to a USB-C power source (5V, 2.0A minimum). Use a standard USB-C cable. A computer USB port or a high-current USB charger is recommended

Communication setup

- Connect to your computer using either USB-C or LAN(Ethernet)
- The device will appear as a COM port for software control.

Reference Input (Optional)

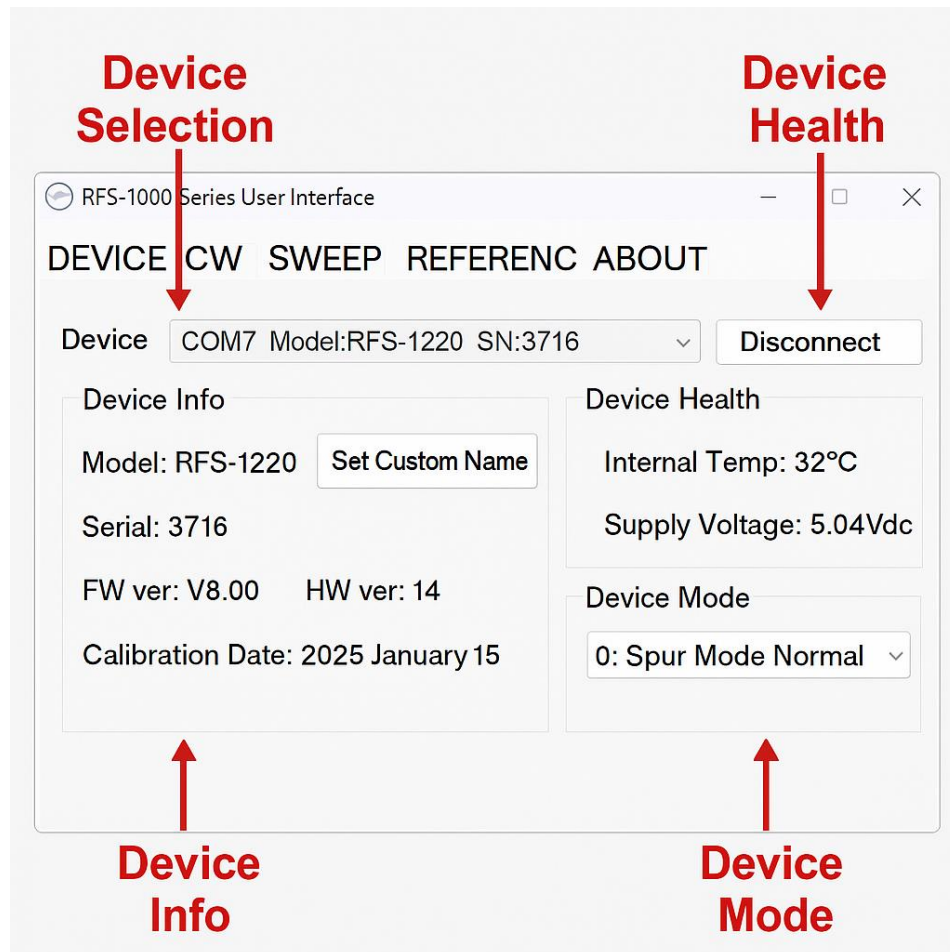
- Connect an external 10 MHz or 100 MHz reference to the REF IN port if preferred. Otherwise, the device will default to its internal OCXO source.

Trigger Setup

- Internal Trigger: select “Internal” in the SWEEP tab of the GUI to start sweeps on-demand by software.
- External Trigger: apply a 5 V TTL pulse to the TRIG connector to initiate the saved sweep (or single-step, if configured) at the rising edge. Always load your sweep parameters first!

Software Interface

- Launch the BNC “RFS-1000 Series User Interface.” (available on the BNC website)
The software will automatically detect and show your connected model, firmware version, and internal temperature.



DEVICE OVERVIEW

Model Variants

- **RFS-1220:** 0.1 – 22 GHz
- **RFS-1420:** 0.1 – 42 GHz

Front Panel Indicators

- 1) CLK IN (10MHz)
- 2) CLK IN (100 MHz)
- 3)RF OUT (SMA)-20GHz
- 4)RF OUT(SMA)-40GHz
- 5)LAN
- 6)POWER
- 7) USB
- 8)TRIGGER (Sweep control)

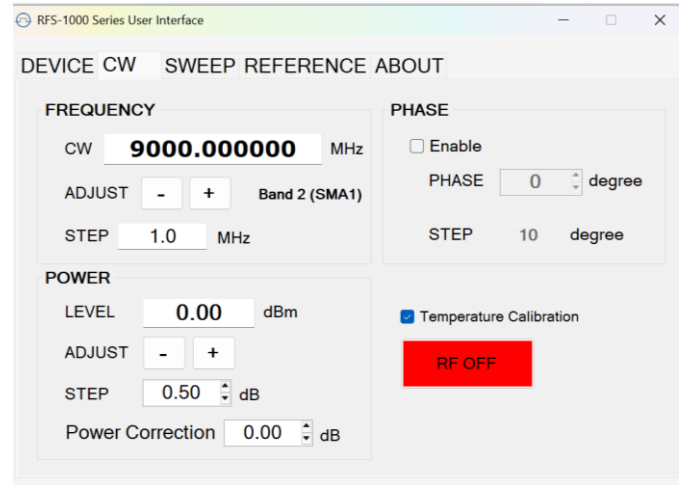


CW Mode Operation

- Navigate to the CW tab in the GUI.
- Set Frequency (in MHz): e.g., 9000.000000
- Adjust Output Power: - Use + / - buttons
- Step size: 0.5 dB
- Enable RF Output: Click the red “RF OFF” button to toggle RF ON

Optional

- Enable and adjust Phase Offset in degrees (0–360°)
- Use “Temperature Calibration” option as needed

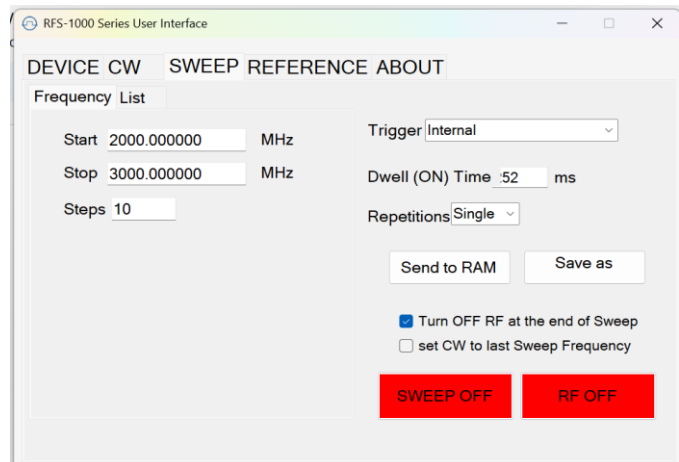


Sweep Mode Operation

- Navigate to the SWEEP tab.
- Set Start/Stop Frequencies: e.g., 2000 MHz to 3000 MHz
- Define Steps (e.g., 10 steps)
- Set Dwell Time (e.g., 52 ms per step)
- Trigger: Internal or external
- Repetitions: Choose Single or Continuous

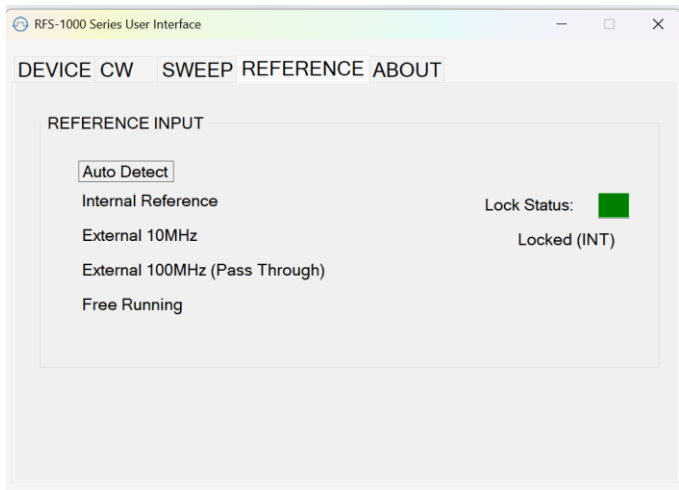
Execution

- Click “Send to RAM” then “SWEEP ON”
- Click “RF ON” to enable RF output
- Optionally enable:
 - “Turn OFF RF at end of sweep” AND “Set CW to last sweep frequency”



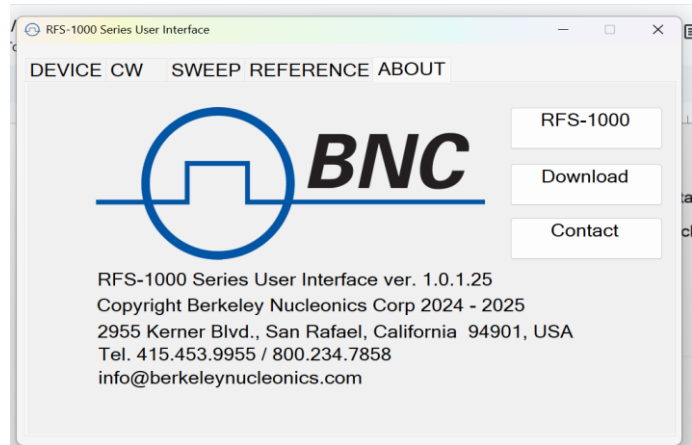
Reference Input Options

- Go to the REFERENCE tab.
- Modes:
 - Auto Detect
 - Internal Reference (OCXO ± 10 ppb)
 - External 10 MHz
 - External 100 MHz (Pass Through)
 - Free Running
- Lock Status: Should show “Locked (INT)” or “Locked (EXT)” for stable operation



ABOUT Tab

- **BNC Logo & Branding**
A large Berkeley Nucleonics Corp. logo centered at the top.
- **Version & Copyright**
Shows the GUI version (e.g. 1.0.1.25) and a copyright line ("Berkeley Nucleonics Corp 2024 – 2025").



Action Buttons

- **RFS-1000:** Opens the product overview or datasheet.
- **Download:** Launches your browser to grab the latest control software.
- **Contact:** Opens your default mail client addressed to BNC support.

SCPI COM COMMAND LIST

Command	Parameters	Comment
*UNITNAME	<string>	Set a unique name in flash memory
*RST		Reset unit now
*DISPLAY	OFF	Power ON or OFF the display
*BUZZER	OFF	Mute the buzzer
*SAVESTATE		Save frequency & attenuation as boot defaults
SWE:MODE	LIST	Select the mode for sweeping (LIST,SCAN)
FREQ:CW	1MHz to 42GHz	Set output Frequency in Hz
OUTP:STAT	1, 0	Turn on or off the RF output
POWER	-40dBm to +10dBm	Set output RF level in dBm
VERNIER 0.6		Set the output power level vernier(fine tune variable attenuator)
SWE:DWELL	100	Sweep dwell time in milliseconds
LIST: DIR	DOWN	Sweep direction

INIT: IMM		Start the sweep now
INIT:CONT		Sweep continuous mode or single
ABORT		Stop the sweep now
FREQ:START	< >	Sweep start frequency in Hz
FREQ:STOP	< >	Sweep stop frequency
SWE:POINTS	<integer>	Sweep point count
LIST:ADD	11GHZ	Add a point to the end of the sweeping list
LIST:CLEAR		Clear the working frequency list and start over
SWE:MODE	OFF	Exits sweeping mode
SWE:MODE	SCAN	Enter scan mode:sweeps between a start point and a stop point
SWE:MODE	LIST	Enter list mode: sweeps an arbitrary list of user frequency points
INIT:IMM		Trigger the sweep now
ABORT		Stop the sweep now
SWE:POINTS		Sets sweep point count
SWE:DWELL		Sweep dwell time in milliseconds
SWE:RESET		Return to the start of the point list(does not clear memory)
FREQ:START		Save the sweep start frequency
FREQ:STOP		Save the sweep stop frequency
INIT:CONT		Sweep continuously or only a single pass
LIST:DIR		Sweep direction
LIST:CLEAR		
LIST:ADD		Add a single point to the end of the sweeping list
INIT:STEPMODE	0	Normal mode where a trigger activates the entire sweep (default)
INIT:STEPMODE	1	Only move forward one point in the sweep

		per trigger activation
INIT:STEPMODE	2	Functions like STEPMODE 1 except that the dwell time is infinite
SWE:ENDRFO		After the seep completes, should RFO state be on or off
SYSREF AUTO		Set REF to Auto Detect
SYSREF FREE		Set REF to Free Running
SYSREF INT		Set REF to Internal Reference
SYSREF EXT		Set REF to Xxternal 10 MHz activation



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